

What is Granularity in Hibernate?

Granularity refers to the **level of detail** at which Hibernate maps Java objects to the database.

Hibernate allows you to choose *how coarse or fine* your mappings should be.

Different annotations enable different levels of granularity.

1. Coarse Granularity

Definition

Hibernate maps a **whole Java class** (entity) to a **single database table**. This is the simplest and least detailed mapping.

Annotation Used

- @Entity
- @Table

Meaning

The entire object is treated as one unit.

Changes to any part of the object trigger persistence for the whole object.

When used?

- Simple domain models
 - No complex structure inside the class
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□2. Medium Granularity (Composite / Component Mapping)

Definition

Hibernate maps **part of an object** (sub-object) into the same table using **embedded components**.

Annotations

- @Embeddable → defines a component class
- @Embedded → used inside the entity

Meaning

You break down an entity into **meaningful components** but store everything in one table.

When used?

- Logical grouping of fields (like Address, Name, Metadata)
 - Reusable components across multiple entities
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3. Fine Granularity (Field-Level Mapping)

Definition

Hibernate maps **individual fields with specific rules**, making the mapping highly detailed.

Common Annotations

- @Column
- @Lob
- @Temporal
- @Enumerated
- @Id, @GeneratedValue

Meaning

You explicitly control how each attribute behaves in the database:

Examples:

- length of a column
- data type conversion
- timestamps
- storing large text or binary data

When used?

- When DB schema requires specific column definitions
 - When fields require custom behavior
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4. Very Fine Granularity (Collection Mapping)

Definition

Hibernate maps **collections of simple values** (not entities) to a **separate table**.

Annotation

- @ElementCollection

Meaning

Each element in the collection becomes a separate row.

This is the most detailed level because Hibernate tracks every individual element.

When used?

- Lists of strings (skills, tags)
 - Sets of numbers (phone numbers, preferences)
 - Maps of simple key-value pairs
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Granularity Comparison Table

Granularity Level	Description	Annotation(s)	Database Structure
Coarse	Entire class mapped as a table	@Entity	1 class → 1 table
Medium	Component inside entity	@Embeddable, @Embedded	1 class + reusable components → 1 table
Fine	Field-specific behavior	@Column, @Temporal, @Lob	Table with detailed column definitions
Very Fin			